

How Weather Affects Transportation

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Purpose

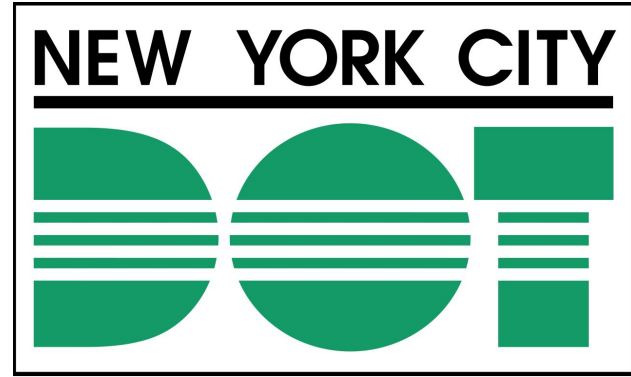
Originally we wanted to observe how different weather conditions affect different types of commute ridership

Now along with that we observe the general trends of how commuting has changed for each borough

Our Stakeholders

The questions we seek to answer through our findings are of a larger scale i.e relating to things such as infrastructure, congestion pricing, etc

As such we believe it would make the most sense if we were to present our findings to the ones who make these decisions, the MTA and DOT



Our General Questions



Where and how should we focus on implementing infrastructure improvements?

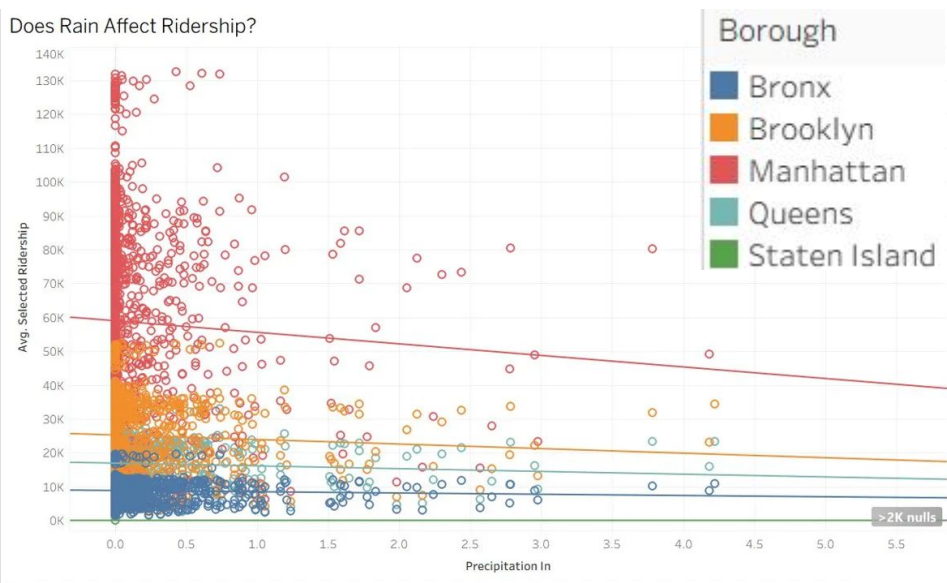
Where should we increase and decrease NYPD presence?

Where and when should we implement congestion pricing?

How Rain Impacts MTA Transit

Subway Ridership

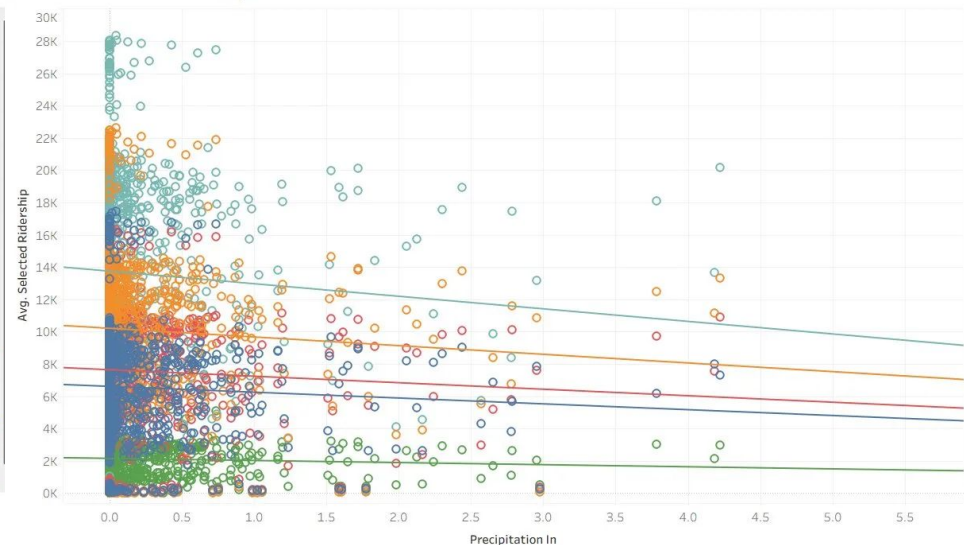
Does Rain Affect Ridership?



We can see subway and bus ridership both decrease for every inch of rain. This tells us MTA transit is significantly impacted for commuters from the rain weather.

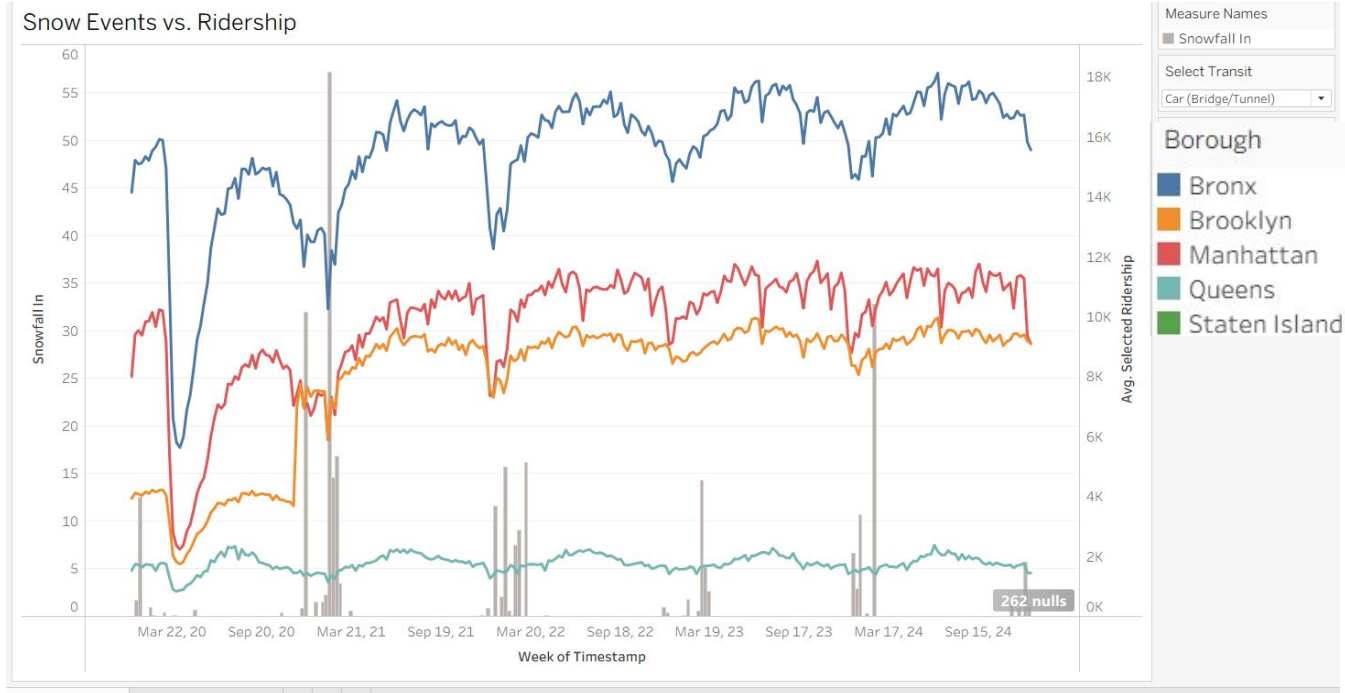
Bus Ridership

Does Rain Affect Ridership?



This becomes an issue when looking at boroughs such as Queens that has no subway access in many areas and relies heavily on bus transit.

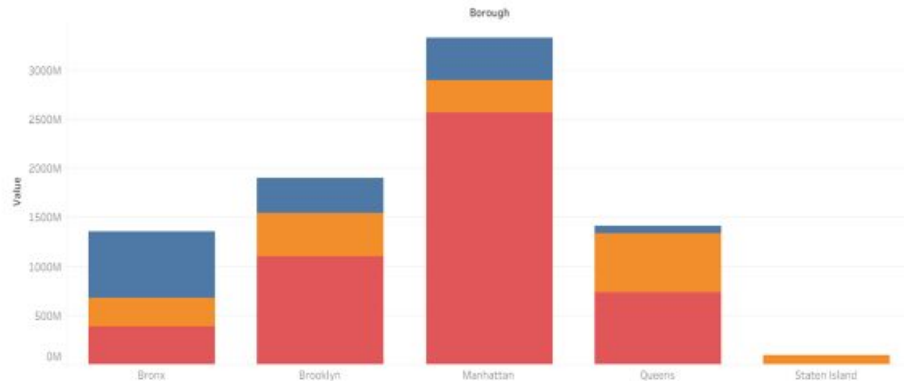
Snow and Car Traffic



Bronx, Manhattan, and Brooklyn have the highest car traffic and snow events cause the largest disruption on roads. This makes these boroughs in most need of weather maintenance to roads and bridges.

Why Infrastructure Needs Differ by Borough

- **Manhattan** — subway-dominant, lowest bus dependency; transit demand is consistent and high
- **Queens** — high bus, near-zero subway; largest mode imbalance across boroughs
- **Bronx & Brooklyn** — high bridge/tunnel volume alongside transit; most car-dependent of the outer boroughs
- **Staten Island** — almost entirely bus + bridge; no subway means any disruption affects the whole borough



Mode dependency determines what kind of infrastructure investment each borough needs most.

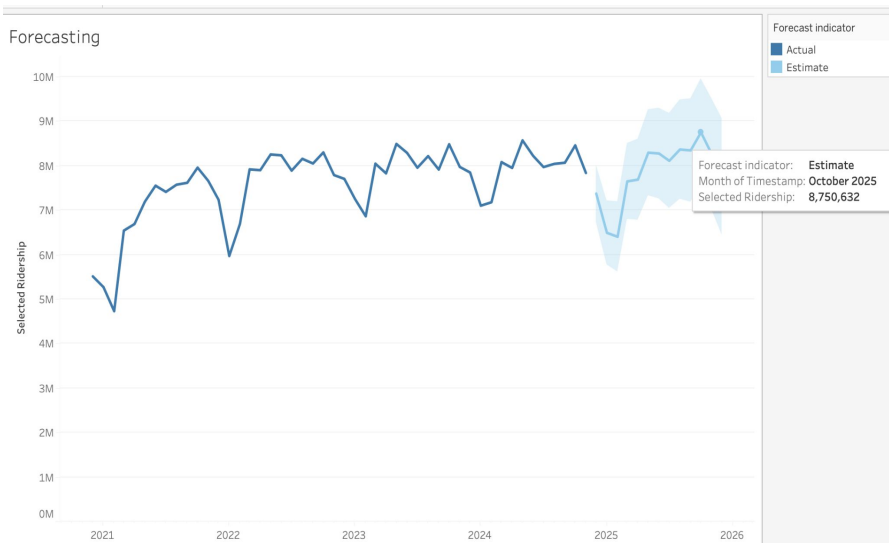
Ridership
Bridge Tunnel Volume
Bus Ridership
Subway Ridership

Infrastructure Investment

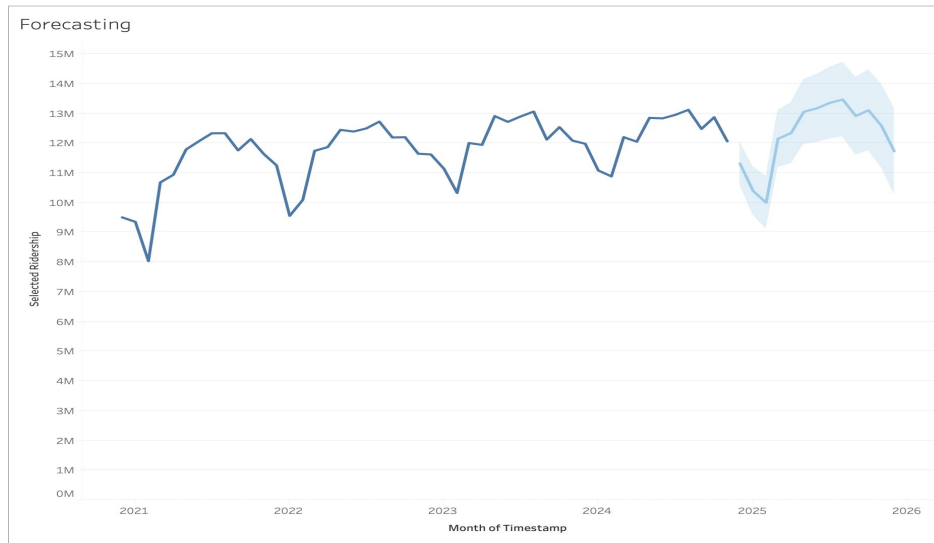
Borough	Problem	Suggestion
Queens	• Bus-dependent • Often no subway fallback	• Subway expansion • Covered bus stops
Staten Island	• Entirely bus and car dependent • Weather impacts all ridership	• Enclosed bus stop shelters • De-icing bridge/roads
Bronx + Brooklyn + Manhattan	• Sharp decrease in traffic during snow events	• Bridge weatherproofing • De-icing roads



Congestion Pricing



This car ridership of manhattan shows its expected to increase and reach its peak in october which seems to be a common trend among other cities to peak in the summer to early fall.



This car ridership of the bronx shows its expected to increase and reach its peak in august and second peak in october. These two cities contain the highest car ridership count which would make them optimal choices to implement congestion pricing.

NYPD Statistics

“an average of 285 cops have quit the force every month since December 2024”

(Dec 2025, New York Post)

- Robbery up by 18%
- Misdemeanor assault up by 15%

(May 2026, New York Post)

NYPD: Time-of-Day Deployment

- **Manhattan** subway ridership is consistently high across all hours. Requires sustained NYPD presence throughout the day, not just rush windows.
- **Brooklyn** shows clear rush hour spikes at 7–9am and 5–7pm. Targeted deployment during those windows is sufficient.
- **Queens and the Bronx** have lower and more even subway ridership. Lighter coverage with a focus on peak hours.

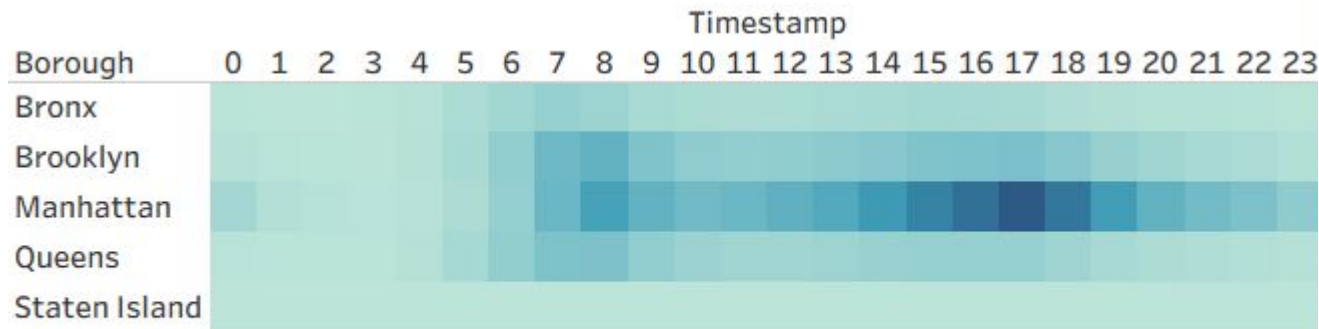


Tableau Public Links

[Weather vs. Transit](#)

[Ridership Forecast](#)

Thank You!

Any Questions?